Development of Improved Acoustic Disdrometer Through Utilization of Machine Learning Algorithm

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Abstract— **Philippines is a tropical country and every year, the country is experiencing typhoons, thunderstorms and excessive rainfalls because of climate change. Since then, the government continuously provides efforts to mitigate natural disasters through numerously growing researches that are inclined with the meteorological processes happening in our country. There are many researches with regards to the methods of quantifying the amount of rainfall but based on those studies, the acoustic disdrometer is rendered useless because of its inability to classify ambient noise from rain types. With this, the main purpose of this study is to develop an improved acoustic disdrometer by adding a capability in which it will categorize the intensity of the amount of rainfall from ambient noise using machine learning algorithm. The proposed methodology is applied by developing a prototype with four piezoelectric sensors, Arduino microcontroller and ZigBee transmitter. Also, the K-nearest neighbors (KNN) predictive model will be established. The obtained results show that the accuracy of the predictive model is 89.95%.**

Keywords— rain gauge, disdrometer, machine learning, KNN, rainfall, climate change

I. INTRODUCTION

In the past decades, climate change has been a great factor in bringing excessive rains in all parts of the world especially in the Philippines, since the rate of evaporation from the ocean is increasing as the world warms [1]. Excessive rainfalls in the Philippines caused numerous cases of flooding and landslides, that took a great number of lives, millions of pesos of properties, infrastructures and livelihood from the Filipino people. Since then, the government continuously provide efforts to mitigate such natural disasters through numerously growing researches that are inclined with the meteorological processes happening in our country, such as weather prediction, flood monitoring and prediction, hydrological availability monitoring [2], and the like. Now that the society is in the age of information technology wherein data can be easily produced, can be readily accessed and can be easily retrieved in a very fast rate unlike before, blooming researches are growing in the field of data science, machine learning and artificial intelligence.

There are many researches with regards to the methods of quantifying the amount of rainfall. Based on these researches, three generalizations were formed: (i) a number of researches tries to continuously improve the accuracy of the tipping bucket; (ii) few researchers make use of acoustic disdrometer because it is merely dependent on sound and it cannot discern whether the data it measures is an actual rain or just an ambient noise; and (iii) disdrometers are paired with radar to improve its accuracy, however, radars are not easily accessed in the country and its components are too expensive. Such observations led to conclude that acoustic disdrometers are rendered useless because of its inability to discern an actual rainfall sound from an ambient noise.

The main objective of this study is to develop an improved acoustic disdrometer by adding a capability in which it will categorize the intensity of the amount of rainfall using a machine learning algorithm. Thus, the specific objectives of the study are: (i) to develop an acoustic disdrometer using piezoelectric sensors; (ii) to use signal processing and machine learning algorithm and its implementation using Python; and (iii) to validate the acquired results.

II. RELATEDWORKS

Traditionally, rain gauge such as tipping bucket is used to measure the amount of rainfall and is subject for improvement. For the past years, studies were done to improve the tipping buckets and one of these is made by Lewlomphaisarl et al. [3] in which they installed an obstacle sheet inside the cone of the tipping bucket to break the cyclone-like behavior of the water and improve its accuracy.

Aside from the tipping bucket, previous studies are done with regards to another device to be used in quantifying the amount of rainfall, and one of these previous attempts is done by Yadnya et al [4] which is all about the measurement of the rain drop distribution in Mataram, India using the acoustic disdrometer for flood prediction. The researchers made use of the acoustic disdrometer to measure the drop size distribution of the rain in order to predict the occurrence of flood along the river of Mataram, based primarily on the two kinds of precipitation – stratiform and convective.

Another study is made by Cao et al [5], which is about the characterization of rain microphysics based on disdrometer and polarimetric radar observation. According to the study, characterization of rain microphysics requires information on raindrop size distributions, however, measurements and retrievals of data contain errors from a single two-dimensional video disdrometer alone so a joint system where a disdrometer and a radar is used.

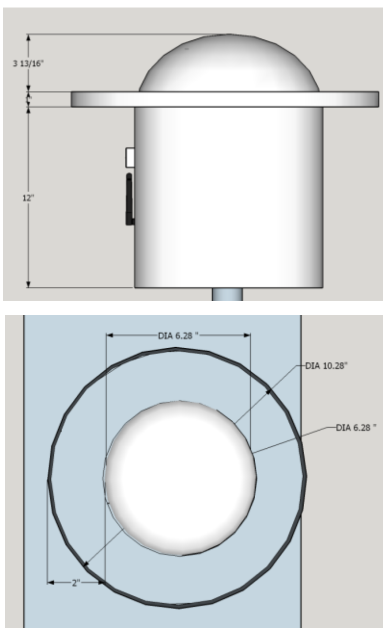
III. Methodology

In this section, the researchers propose a systematic plan, through methods and techniques, on how to categorize the amount of rainfall using machine learning algorithm.

A. Hardware or Prototype Development

The first part of the study was the hardware development of the acoustic disdrometer. The acoustic disdrometer is a cylindrical device constructed by integrating piezoelectric sensors, readout circuit, and Arduino Uno microcontroller. Data transmission was either wireless via Zigbee transmitter or manually retrieved via SD card inside an SD card module connected in the microcontroller. Moreover, the prototype got its supply from a power bank installed in the body or be connected via USB connector when needed to be charged.

The design and measurement of the device was compliant in terms of the WMO standards on precipitation gauges [6]. Figure 1 shows the dimensions of the prototype.



(a)

(b)

Fig. 1. (a) Cylindrical body of disdrometer, side view; (b) Dome and catch basin of disdrometer, top view.

TABLE 1. Data Gathering Scheme.

Rainfall Classification	Rainfall Measurement
Light	Less than 2.5 mm or 2.5 liters per square meter per hour
Moderate	2.5 to 7.5 mm or 2.5 to 7.5 liters per square meter per hour
Heavy	7.5 to 15 mm or 7.5 to 15 liters per square meter per hour
Intense	15 to 30 mm or 15 to 30 liters per square meter per hour
Torrential	More than 30 mm or 30 liters per square meter per hour

B. Data Gathering

The proposed data gathering scheme involved quantifying the amount of rainfall using four piezoelectric sensors. The source of data was from a rainfall simulator which established the five classifications of rainfall, namely – light, moderate, heavy, intense and torrential.

The rainfall simulator and the acoustic disdrometer were both placed in an open ground, but the acoustic disdrometer was mounted 0.8 m above the ground based on the WMO standards [6]. The acoustic disdrometer recorded the data obtained from the four piezoelectric sensors, for every second, in a comma-separated values (CSV) file which was both stored in an SD card and was transmitted wirelessly via ZigBee transmitter [7, 8]. The acoustic disdrometer was exposed to the different ambient noises and the different levels of rain produced by the rainfall simulator, for about 25 minutes for each kind of sound, producing a total of six csv files or a total of 9,000 samples.

The different levels of rain produced by the rainfall simulator was based on [9] set by DOST-PAGASA. Table 1 shows the data gathering scheme used in the study.

The six csv files were imported in the program wherein the piezoelectric sensor readings were used as predictors and the rainfall classification were set as the target in the dataset.

After retrieving the dataset from the setup, the data undergo data cleaning and data tidying wherein the data frame was organized, with each variable forming a column, each sample forming a row, and each type of observational unit forming a table [10].

Afterwards, the dataset underwent data slicing as a pre-requisite in making a training set. In this section, the dataset was sampled and divided into two parts: the training data, which was used to build predictive models, and the test data, which was used in validating the acquired results. In the study, the researchers adopted the standard partition for the data slicing which was, 70% for the training data and 30% for the test data.

C. Development of Predictive Model using KNN

K-nearest neighbors (KNN), a machine learning algorithm under supervised learning, was used to create the predictive model for the dataset attained from the previous section.

1) Model training and algorithm tuning. KNN algorithm is one of the top 10 most significant algorithms in data mining [11]. It is a non-parametric machine learning algorithm that uses both nominal and numerical characteristics of data through the selection of the most common characteristic between the k neighbors. Figure 2 shows the implementation of KNN as derived from [12].

For validation, the researchers used the standard cross-validation method which was the 10-fold 10 repeats, as in [13], in evaluating the performance of the KNN predictive model.

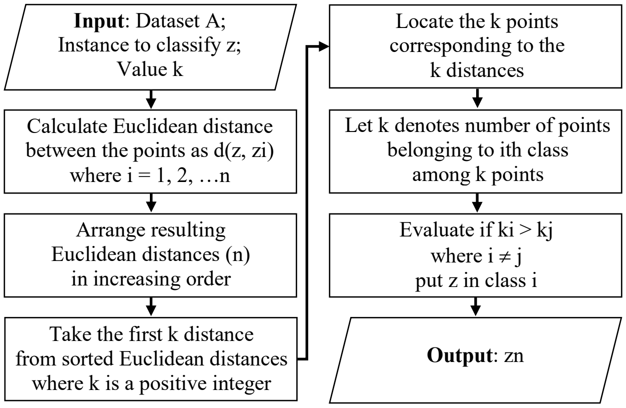


Fig. 2. Flowchart for KNN pseudocode.

2) Model evaluation. After building and validating the predictive model, the model was evaluated in terms of accuracy. Equation 1 shows the formula used in calculating the accuracy where TP stands for true positive, TN stands for true negative, FP stands for false positive, and FN stands for false negative.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \qquad (1)$$

IV. Results and Discussion

This section presents the interpretations of the results of the process and techniques involved.

A. Project Prototype

The actual prototype of the acoustic disdrometer is shown in Figure 3. The Arduino microcontroller is the main processing unit in the device since it records the sensor readings in an SD card through an SD card module and controls the ZigBee transmitter for the wireless data transmission.

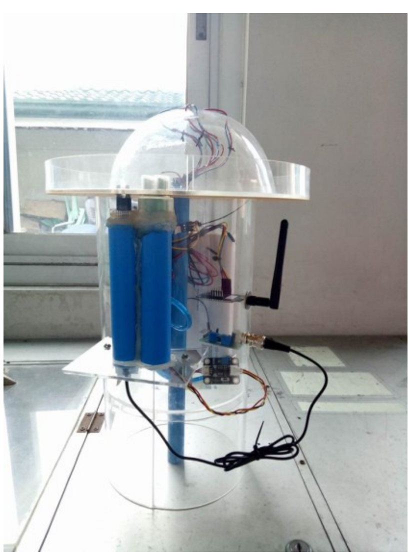


Fig. 3. Photo of improved acoustic disdrometer protoype.

B. Exploratory Data Analysis

The boxplot diagram is a standardized way of displaying the shape of the distribution and determines whether the data needs centralization due to scattered points in the graph. Figure 4 shows the boxplot diagram of the piezoelectric sensors reading for each rain type. Based from the boxplot diagram, the dots, which signified the other samples, were outside the box which implied that the dataset needed normalization. Normalization is the center at scale of the code and currently implemented to make the data structures comparable. The dataset obtained from Piezoelectric Sensor 1 had the widest range of concentrations compared to the other three piezoelectric sensors.

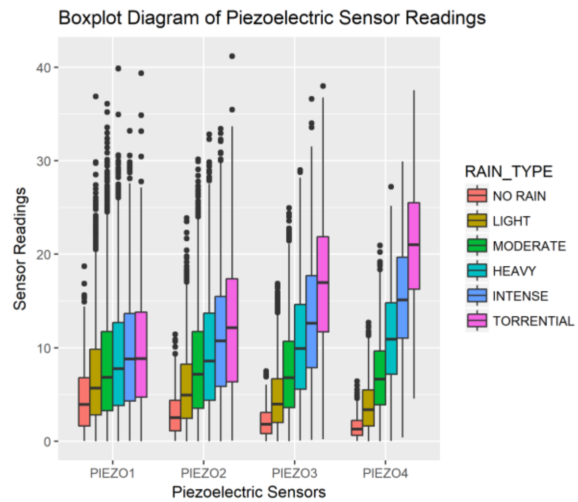


Fig. 4. Boxplot diagram of piezoelectric sensor readings

In implementing machine learning algorithms, it is needed for the data to be correlated, thus, a total of six sets of correlation graphs for comparing the sensor readings of the four piezoelectric sensors were generated. For the visualization purposes, since the sets of graphs were almost identical, the correlation graph of piezoelectric sensors 3 and 4 sensor readings is the only set of graphs presented in this paper. It can be seen from the graph in Figure 5 that the sensor readings were linearly correlated for each rain type.

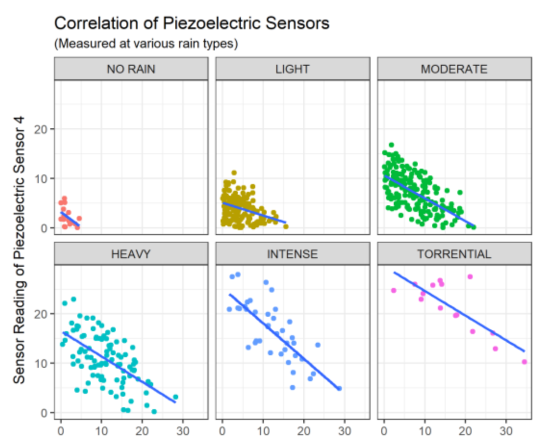


Fig. 5. Correlation of Piezoelectric Sensors 3 and 4 at various rain types

C. KNN Implementation Results

Figure 6 shows the tuning graph of the predictive model. Based from the tuning results of KNN predictive model using the standard partition of 10-folds, 10-repeats, repeated cross-validation, the accuracy of the predictive model occurred at a value of 0.8995.

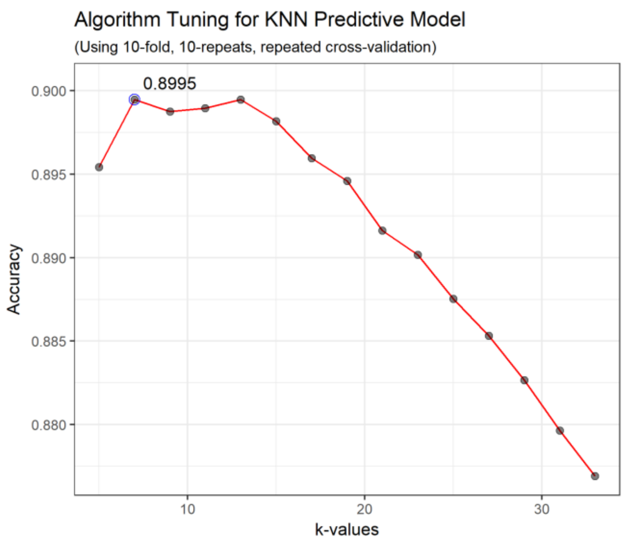


Fig. 6. Algorithm Tuning for KNN Predictive Model

Table 2 gives the confusion matrix corresponding to the overall statistics of the KNN predictive model. Indicated in the table are the different types of rainfall (RF), starting from no rainfall condition (N), to light rainfall (L), to moderate rainfall (M), to heavy rainfall (H), to intense rainfall (I), and to torrential rainfall (T).

TABLE 2. CONFUSION MATRIX

RF	N	L	M	H	I	T
N	91	11	0	0	0	0
L	11	647	40	0	0	0
M	0	36	887	59	0	0
H	0	0	38	558	45	0
I	0	0	0	11	182	25
T	0	0	0	0	1	56

V. CONCLUSION

Based from the obtained data results, it can be concluded that the development of acoustic disdrometer using K-nearest neighbors (KNN) as machine learning predictive model was implemented successfully with an accuracy of 89.95%. The types of rainfall considered in the predictive model included the no rainfall, light, moderate, heavy, intense and torrential rainfalls.

ACKNOWLEDGMENT

This work is related with the project funded by the National Water Resources Board (NWRB) and by the Philippine Council for Industry, Energy and Emerging Technology Research and Development of the Department of Science and Technology (DOST-PCIEERD) with Project No. 3982 and year of Project Start 2016.

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